

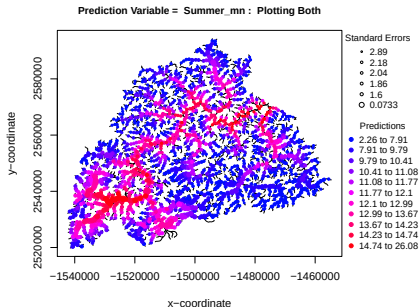
Spatial Statistical Models for Stream Networks

Jay Ver Hoef

National Marine Mammal Lab
NOAA Fisheries Alaska Science Center
International Arctic Research Center
Fairbanks, Alaska, USA

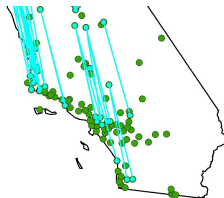
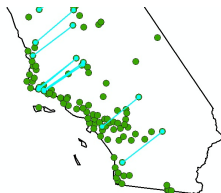
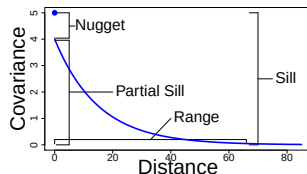
Outline

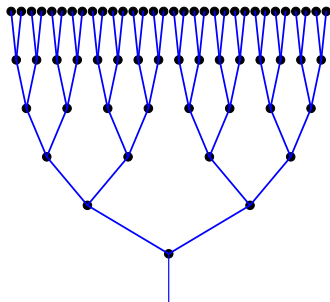
- ▶ Introduction
- ▶ Moving Average Models
- ▶ ESDA
- ▶ Prediction
- ▶ Regression
- ▶ Sampling
- ▶ Sample Designs
- ▶ Summary



Reduce the number of parameters in the probability model by using spatial relationships

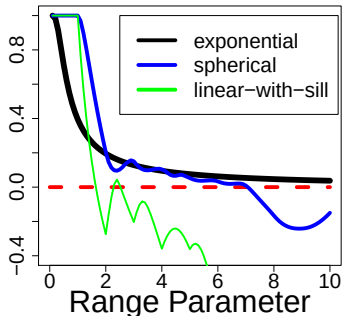
$$\Sigma = \begin{pmatrix} \sigma_{1,1} & \sigma_{1,2} & \cdots & \sigma_{1,n} \\ \sigma_{2,1} & \sigma_{2,2} & \cdots & \sigma_{2,n} \\ \vdots & \vdots & \cdots & \vdots \\ \sigma_{n,1} & \sigma_{n,2} & \cdots & \sigma_{n,n} \end{pmatrix}$$



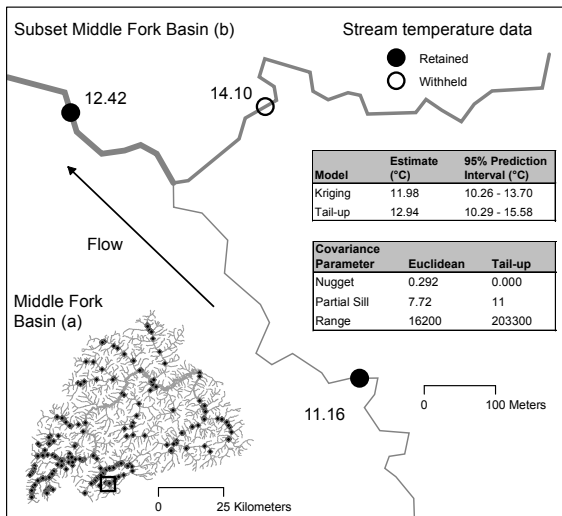


Not to scale: all lengths 1
unit

Minimum Eigenvalue

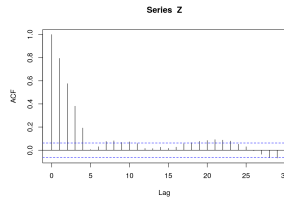
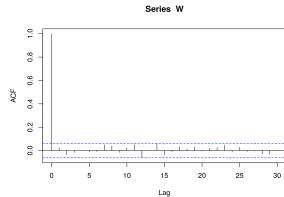
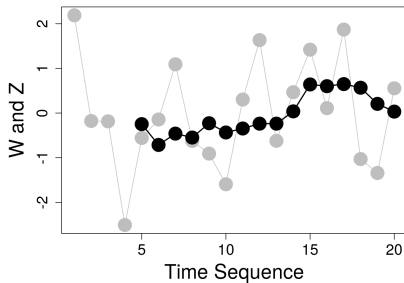


Why Use Stream Network Models?

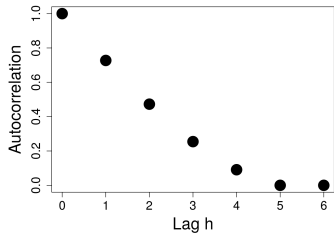
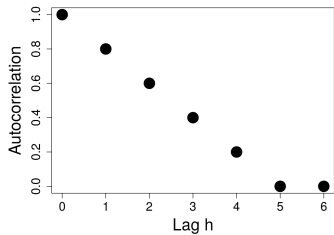
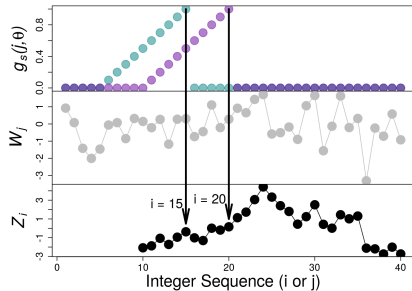


Discrete Moving Average Construction

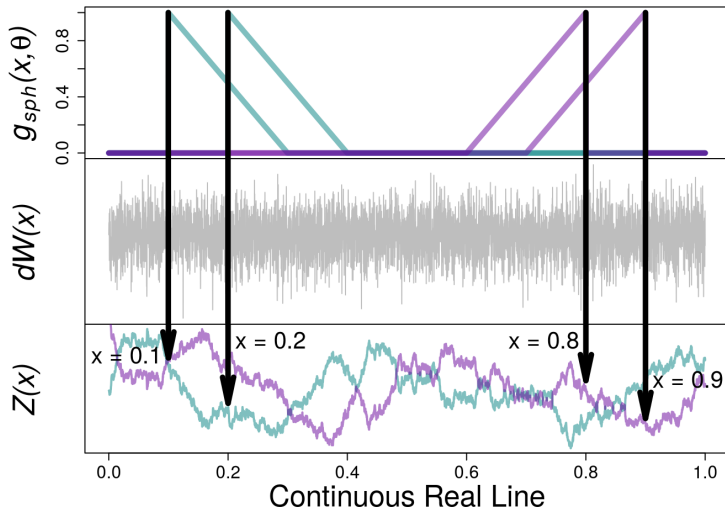
$$Z_i = \sum_{k=i-\theta+1}^i W_k = W_i$$



Discrete Moving Average Construction

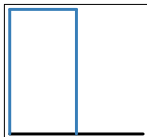


Continuous Moving Average Construction

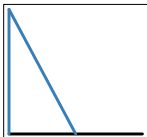


Autocorrelation from Moving Average Construction

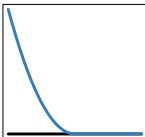
Moving Average Function



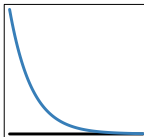
Moving Average Function



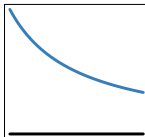
Moving Average Function



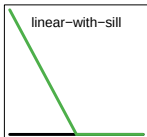
Moving Average Function



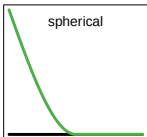
Moving Average Function



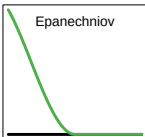
Autocorrelation Function



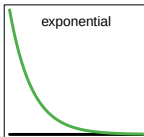
Autocorrelation Function



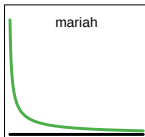
Autocorrelation Function



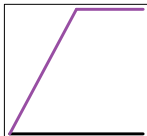
Autocorrelation Function



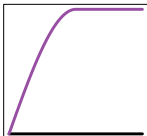
Autocorrelation Function



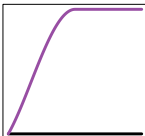
Semivariogram Function



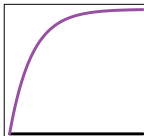
Semivariogram Function



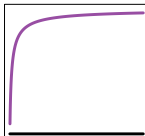
Semivariogram Function



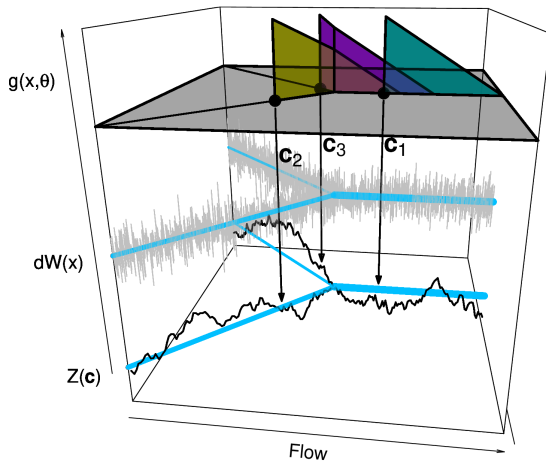
Semivariogram Function



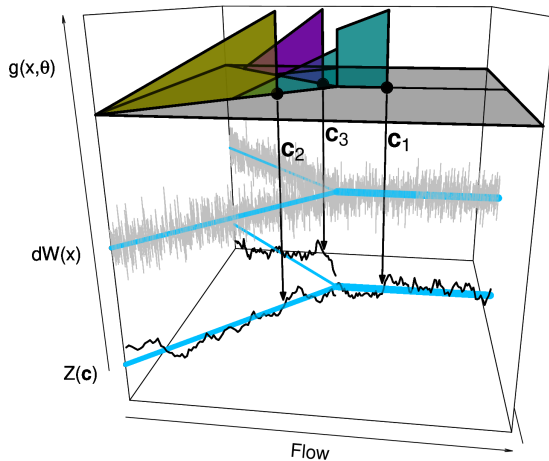
Semivariogram Function



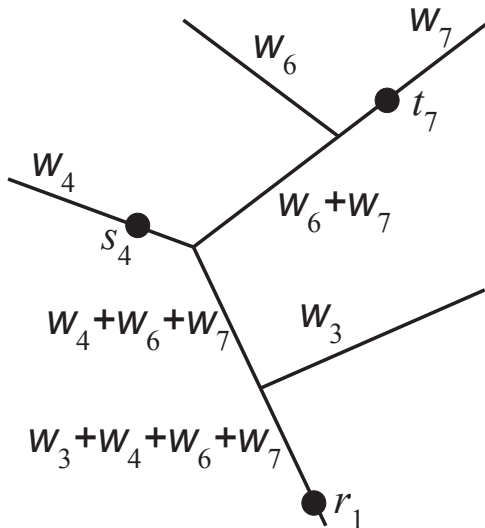
Tail-Down Models from Moving Average Construction



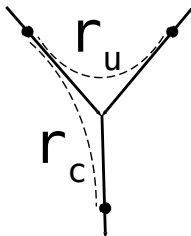
Tail-up Models from Moving Average Construction



Weighting for Additive Function Values



Type	Model	Autocorrelation
Tail-up	All	Autocorrelation only when flow-connected $r_u = 0$, $0 < r_c < 1$
Mixed	Component	More autocorrelation between flow-connected $r_c \geq r_u$
Tail-down	Exponential	Equal autocorrelation $r_c = r_u$
Tail-down	All Others	More autocorrelation between flow-unconnected $r_c \leq r_u$



Linear Mixed Model

$$Y(s) = \mathbf{x}(s)' \beta + Z_{Up}(s) + Z_{Down}(s) + Z_{Euc}(s) + Z_{RE} + \epsilon(s)$$
$$\Sigma(\theta) = \text{var}(\mathbf{z}_{Up} + \mathbf{z}_{Down}(s) + \mathbf{z}_{Euc}(s) + \mathbf{z}_{RE} + \epsilon)$$

Fit with **Maximum Likelihood** or **Restricted Maximum Likelihood (REML)** for regression parameters (fixed effects) β and covariance (autocorrelation) parameters θ

Why REML?

Consider simple i.i.d. mean-only model

$$\hat{\sigma}_{ml}^2 = \frac{\text{biased}}{n} \sum_{i=1}^n (y_i - \bar{y})^2 \quad \hat{\sigma}_{reml}^2 = \frac{\text{unbiased}}{n-1} \sum_{i=1}^n (y_i - \bar{y})^2$$

Selection Methods

1. stepwise forward selection
2. stepwise backward removal (+ re-entry)
3. all subsets
4. just a few candidate models

Selection Criteria

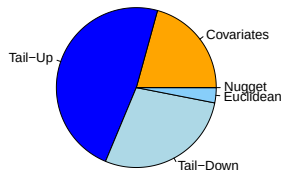
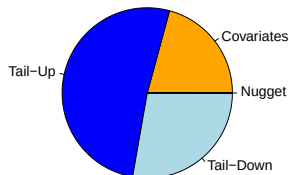
1. P-values (only useful for explanatory variables)
2. AIC, BIC, other Information criteria based on likelihood
3. leave-one-out crossvalidation (LOOCV)

Some Rules

1. Maximum likelihood can always be used
2. Restricted maximum likelihood (REML) should NOT be used for likelihood-based criteria if explanatory variables are different among models

$$Y(s) = \beta_0 + \beta_1 \text{Ele} + \beta_2 \text{Slp} + Z_{up}(s) + Z_{dn}(s) + Z_{Eu}(s) + \epsilon(s)$$

Model	Type	AIC	RMSPE	cov.90
IND	IND	405.39	2.41	0.99
LWS	UP	232.27	2.11	0.86
EXP,EXP	UP,DN	230.91	1.88	0.89
EXP,EXP,EXP	UP,DN,EU	243.26	1.88	0.89
LWS	DN	305.62	2.01	0.88



$$Y(s) = \beta_0 + \beta_1 \text{Ele} + \beta_2 \text{Slp} + Z_{up}(s) + Z_{dn}(s) + Z_{Eu}(s) + \epsilon(s)$$

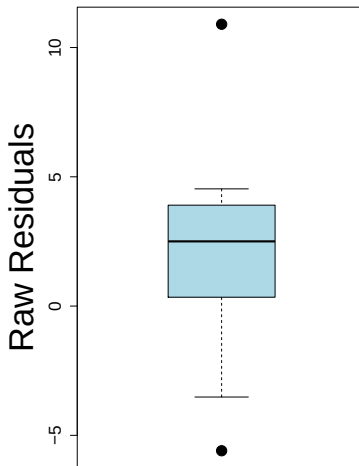
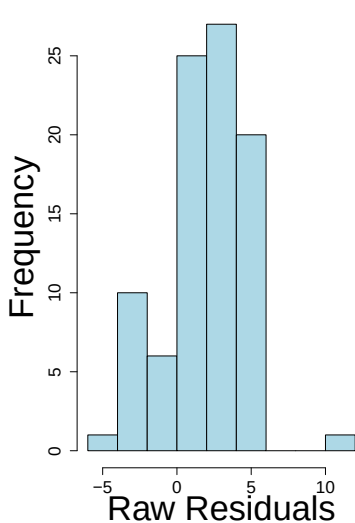
Table: Tail-Up + Tail-Down

Factor	Estimate	std.err	t.value	prob.t
β_0	44.342	7.415	5.980	<0.0001
β_1	-0.017	0.004	-4.698	<0.0001
β_2	-7.497	13.947	-0.538	0.5923

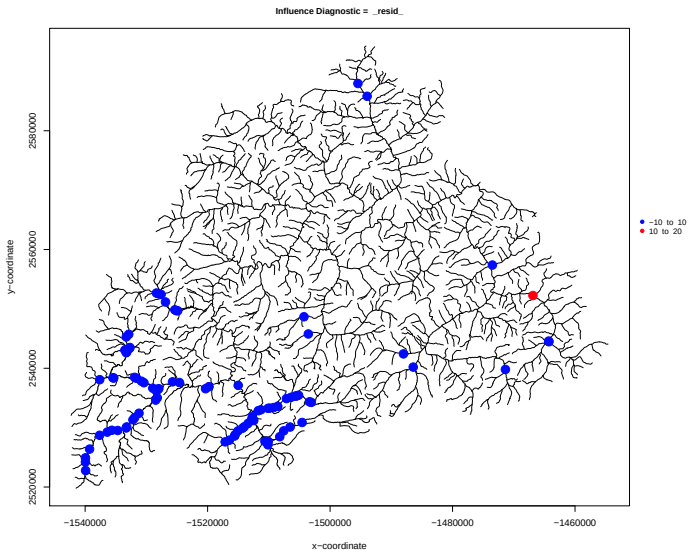
Table: Independence

Factor	Estimate	std.err	t.value	prob.t
β_0	29.429	5.177	5.685	<0.0001
β_1	-0.008	0.003	-3.217	0.0018
β_2	-73.276	22.061	-3.322	0.0013

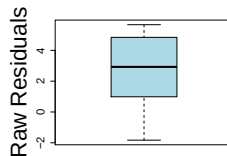
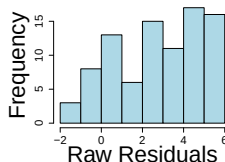
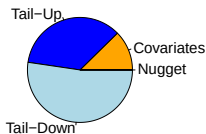
Diagnostics - Histogram and Boxplot



Diagnostics - Mapping Outlier



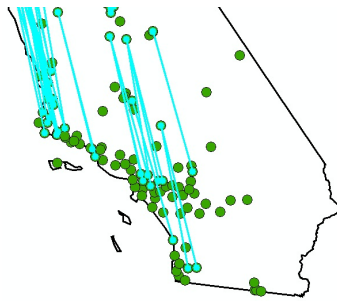
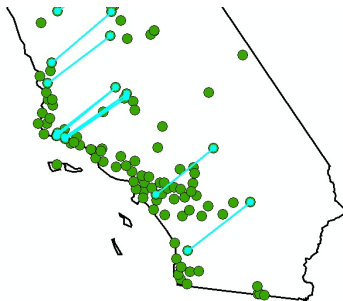
Factor	Estimate	std.err	t.value	prob.t
(Intercept)	29.541	5.888	5.017	0.0000
ELEV	-0.010	0.003	-3.501	0.0007



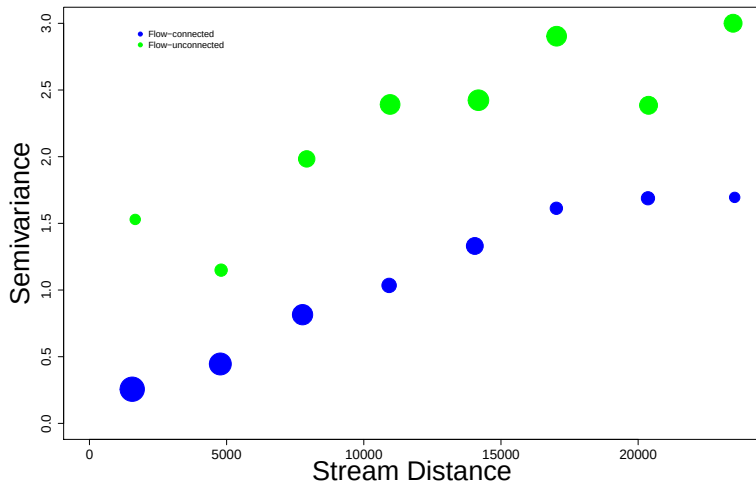
semivariogram(distance h) =

$$\frac{1}{2} \text{ average}[(\text{value at location } i - \text{value at location } j)^2]$$

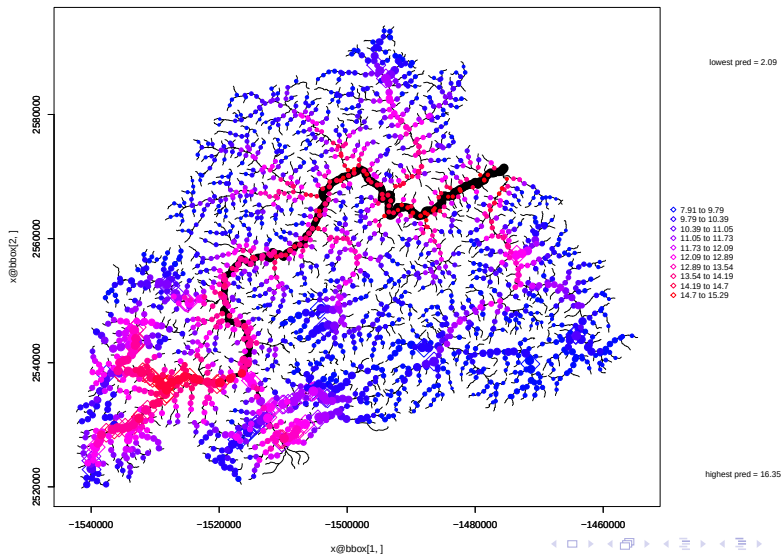
for all pairs of locations i and j separated by distance h .



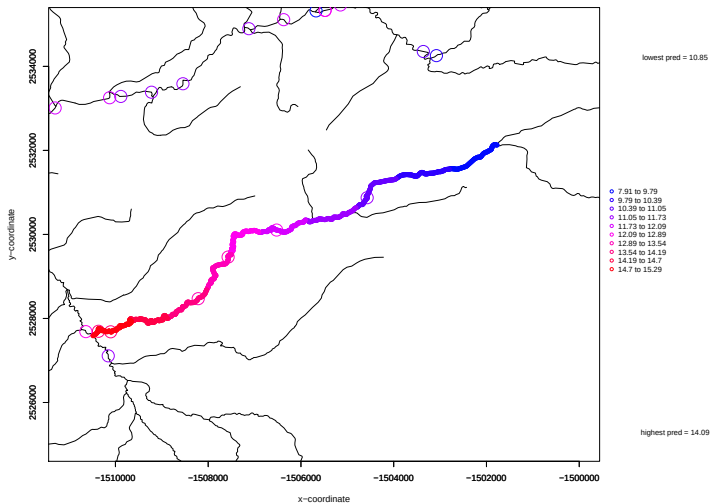
Residual Analysis



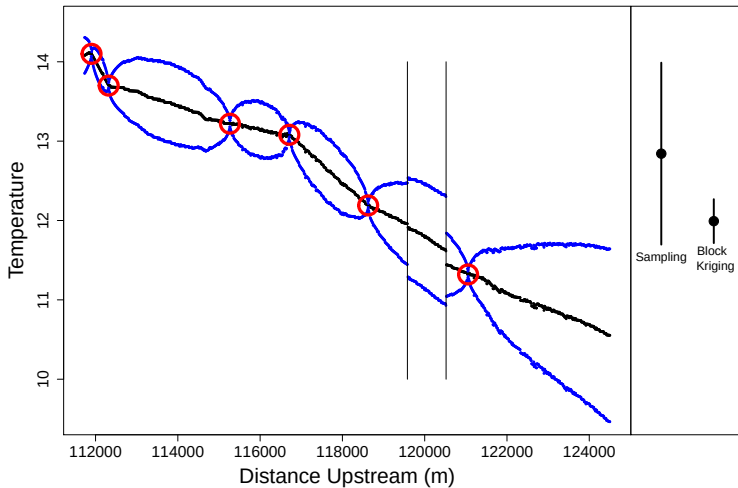
SLM Prediction Map



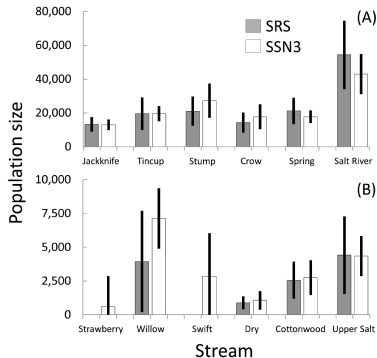
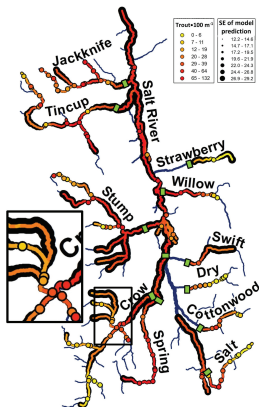
Dense Grid of Points



Block Kriging



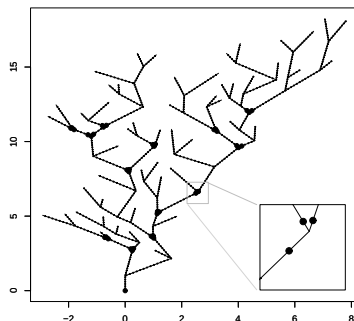
Block Kriging



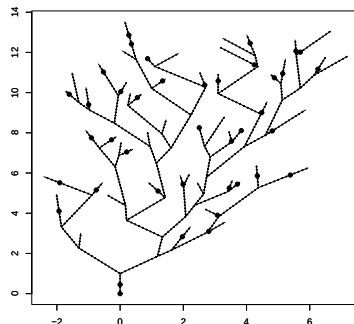
SRS Total: $155,828 \pm 26,514$

SSN Total: $184,030 \pm 27,263$

EK Criteria - Tail Up Model



EK Criteria - Tail Up + Down



- ▶ **Ver Hoef, J.M., Peterson, E.E.,** and Theobald, D. 2006. Spatial statistical models that use flow and stream distance. *Environmental and Ecological Statistics* **13**: 449-464.
- ▶ **Peterson, E.E.,** Theobald, D.M., and **Ver Hoef, J.M.** 2007. Geostatistical modelling on stream networks: Developing valid covariance matrices based on hydrologic distance and stream flow. *Freshwater Biology* **52**: 267-279.
- ▶ Garreta, V., Monestiez, P. and **Ver Hoef, J.M.** 2010. Spatial modelling and prediction on river networks: Up model, down model or both? *Environmetrics* **21**: 439-456.
- ▶ **Ver Hoef, J.M.** and **Peterson, E.E.** 2010. A moving average approach for spatial statistical models of stream networks (with discussion). *Journal of the American Statistical Association* **105**: 6-18.
- ▶ **Peterson, E.E.** and **Ver Hoef, J.M.** 2010. A mixed-model moving-average approach to geostatistical modeling in stream networks. *Ecology* **91(3)**: 644-651.
- ▶ **Peterson, E.E., Ver Hoef, J.M., Isaak, D.J.,** Falke, J.A., Fortin, M-J, Jordan, C., McNyset, K., Monestiez, P., Ruesch, A.S., Sengupta, A., Som, N., Steel, A., Theobald, D.M., Torgersen, C.E., Wenger, S.J. 2013. Stream networks in space: concepts, models, and synthesis. *Ecology Letters* doi: 10.1111/ele.12084.
- ▶ **Ver Hoef, J.M., Peterson, E.E.,** Clifford, D. and Shah, R. 2014. SSN: An R Package for Spatial Statistical Modeling on Stream Networks. *Journal of Statistical Software* **56(3)**: 1-45.
- ▶ **Peterson, E.E.** and **Ver Hoef, J.M.** 2014. STARS: An ArcGIS Toolset Used to Calculate the Spatial Information Needed to Fit Spatial Statistical Models to Stream Network Data. *Journal of Statistical Software* **56(2)**: 1-17.
- ▶ **Isaak, D.J., Peterson, E.E., Ver Hoef, J.M.,** Wenger, S.J., Falke, J.A., Torgersen, C.E., Sowder, C., Steel, E.A., Fortin, M-J, Jordan, C.E., Ruesch, A.S., Som, N. and Monestiez, P. 2014. Applications of Spatial Statistical Network Models to Stream Data. *WIREs Water* **1(3)**: 279-294. doi: 10.1002/wat2.1023.

- ▶ Som, N.A., Monestiez, P., **Ver Hoef, J.M.**, Zimmerman, D.L. and **Peterson, E.E.** 2014. Spatial Sampling on Streams: Principles for Inference on Aquatic Networks. *Environmetrics* **25(5)**: 306-323. DOI: 10.1002/env.2284.
- ▶ Rushworth, A.M., **Peterson, E.E.**, **Ver Hoef, J.M.** and Bowman, A.W. 2015. Validation and Comparison of Geostatistical and Spline Models for Spatial Stream Networks. *Environmetrics* **26(5)**: 327–338. DOI: 10.1002/env.2340.
- ▶ **Isaak, D.J.**, Young, M.K., Luce, C.H., Hostetler, S.W., Wenger, S.J., **Peterson, E.E.**, **Ver Hoef, J.M.**, Groce, M., Horan, D.L. and Nagel, D. 2016. Slow Climate Velocities of Mountain Streams Portend their Role as Refugia for Cold-Water Biodiversity. *Proceedings of the National Academy of Sciences* **113**: 4374–4379.
- ▶ **Isaak, D.J.**, **Ver Hoef, J.M.**, **Peterson, E.E.**, Horan, D., and Nagel, D. 2017. Scalable Population Estimates Using Spatial-Stream-Network (SSN) Models, Fish Density Surveys, and National Geospatial Database Frameworks for Streams. *Canadian Journal of Fisheries and Aquatic Sciences* **74(2)**: 147–156.
- ▶ Zimmerman, D.L. and **Ver Hoef, J.M.** 2017. The Torgegram for Fluvial Variography: Characterizing Spatial Dependence on Stream Networks. *Journal of Computational and Graphical Statistics* **26:2** 253–265.
- ▶ **Isaak, D.J.**, Wenger, S.J., **Peterson, E.E.**, **Ver Hoef, J.M.**, Nagel, D.E., Luce, C.H., Hostetler, S.W., Dunham, J.B., Roper, B.B., Wollrab, S.P., Chandler, G.L., Horan, D.L., Parkes-Payne, S. 2017. The NorWeST Database and Modeled Summer Temperature Scenarios: Massive Crowd-sourcing and New Geospatial Tools Reveal Broad Climate Warming of Rivers and Streams in the Western U.S. *Water Resources Research*: **53(11)**: 9181–9205. DOI: 10.1002/2017WR020969.
- ▶ **Ver Hoef, J.M.** 2018. Kriging Models for Linear Networks and non-Euclidean Distances: Cautions and Solutions. *Methods in Ecology and Evolution* DOI: 10.1111/2041-210X.12979
- ▶ **Ver Hoef, J.M.**, **Peterson, E. E.**, **Isaak, D. J.** Review of Spatial Statistical Models for Stream Networks, In press in Gelfand, A., Fuentes, M., Hoeting, J., and Smith, R. (eds.), *Handbook of Environmental and Ecological Statistics*, Chapman and Hall.